

Fig. 2: An actual-size, single-side PCB for the mains monitor

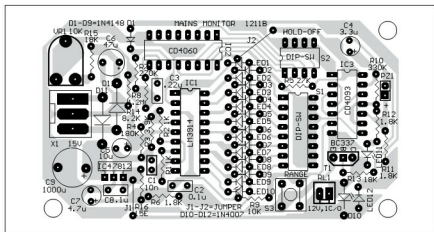


Fig. 3: Component layout for the PCB

age range.

Assemble the circuit on a PCB as it minimises time and assembly errors. An actual-size, single-side PCB for the mains monitor is shown in Fig. 2 and its component layout in Fig. 3. Carefully assemble the components and double-check for any overlooked error.

For testing the circuit, vary the AC power to the mains monitor circuit using an auto transformer. Set preset VR1 to its middle position, hold-off time to 15 seconds using DIP switch S2 and voltage range to one step on either side of 240V AC (235V and 245V) using DIP switch S1. LED6 should glow to indicate that 240V AC is selected as you vary the AC power using the auto transformer. If it does not glow, adjust preset VR1.

Assuming that mains voltage is within the acceptable range, relay RL1 energises, LED11 glows and the load connects to AC mains 15 seconds after powering the circuit. As you vary the auto transformer output to an unacceptable value, say, 230V AC, LED4 glows, relay RL1 de-energises, LED12 flashes and the load disconnects from mains supply. LED11 glows continuously when you press memory reset switch S3. ●

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